6 DOF Aircraft Simulation in MATLAB Simulink

Overview

This repository contains a complete 6 Degrees of Freedom (6 DOF) aircraft dynamics simulation built using MATLAB Simulink. The project models the full aircraft dynamics including control inputs such as elevator, rudder, aileron, and dual-engine throttle. The simulation provides a comprehensive set of outputs, including:

- Linear Velocities in Body Frame (u, v, w)
- Euler Angles (Roll, Pitch, Yaw)
- Angular Rates (p, q, r)
- Velocities in the NED (North-East-Down) Frame
- Position in NED Frame (x, y, z)
- Geodetic Position (Latitude, Longitude, Height)

The aircraft motion and maneuvers are visualized using the MATLAB Aircraft Animation Toolbox.

Repository Contents

- `rcam.m` - Function file used in the Simulink model to define the aircraft dynamics
- `initialization_integrator_constant.m` - Initialization script for constants and simulation setup
- Simulink Model File (`.slx`) - Integrated Simulink model using the above components
- Aircraft Animation Toolbox files - Required for visualizing aircraft motion
- Supporting functions and data files for coordinate transformation and geodetic calculations

2. Requirements

- MATLAB R2020a or later
- Simulink
- Aerospace Toolbox (for animation and coordinate conversion)

3. Running the Simulation

1. Download or clone the repository and open it in MATLAB.
2. Open and run `initialization_integrator_constant.m` to initialize the simulation parameters.
3. This script also launches the main Simulink model and sets the initial conditions.
4. Ensure `rcam.m` is in the MATLAB path since it is called by the Simulink model.

Initial Conditions

The following initial conditions are defined in the script. You may modify them as per your requirements.

State Vector (`x0`)

```
x0 = [85;
      0;
      0;
      0;
      0;
      0;
      0;
      0;
      0];
```

Initial Velocity in Body Frame (`v0`)

```
v0 = [0;
      0;
      0];
```

Control Inputs (`u`)

```
u = [0;         % Aileron
    -0.1;       % Elevator (-6 degrees)
     0.05;      % Rudder
     0.08;      % Right Wing Throttle
     0.08];     % Left Wing Throttle
```

Features

- Full nonlinear 6 DOF aircraft simulation using integrated equations of motion
- Dual-engine input modeling
- Euler angle and rate calculation
- Velocity transformation into the NED coordinate frame
- Geodetic (Lat, Lon, Height) position computation
- Real-time aircraft maneuver visualization using Animation Toolbox

Credits

This simulation was developed as a dynamic model of aircraft motion with control system inputs and real-time visualization using MATLAB Simulink and the Aircraft Animation Toolbox.

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